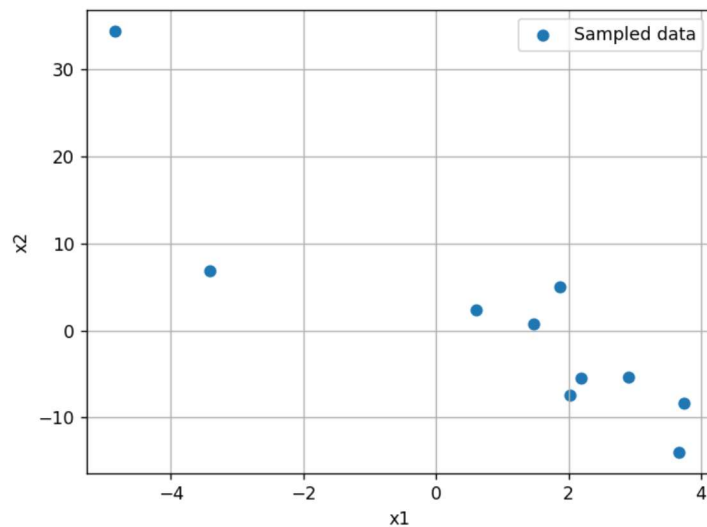
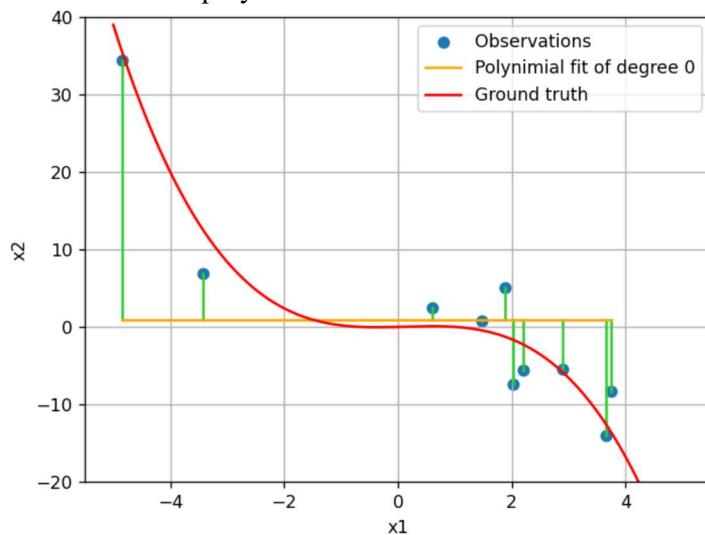


Exercise 1

Plot of raw data:

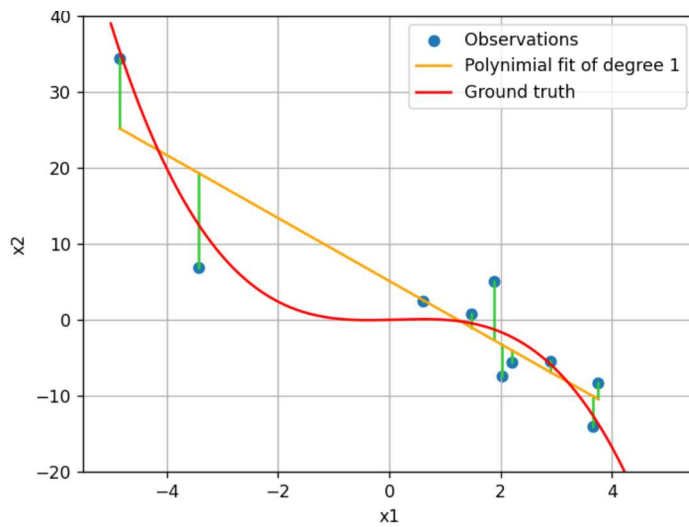


Plots of various polynomials:



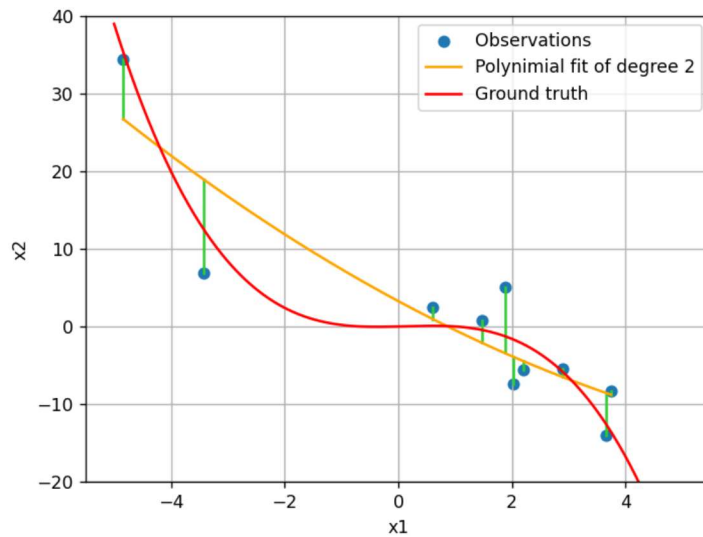
MSE=163.883

A 0-degree polynomial tells us nothing else than the average value of the selected data, which in this case is negligent information.



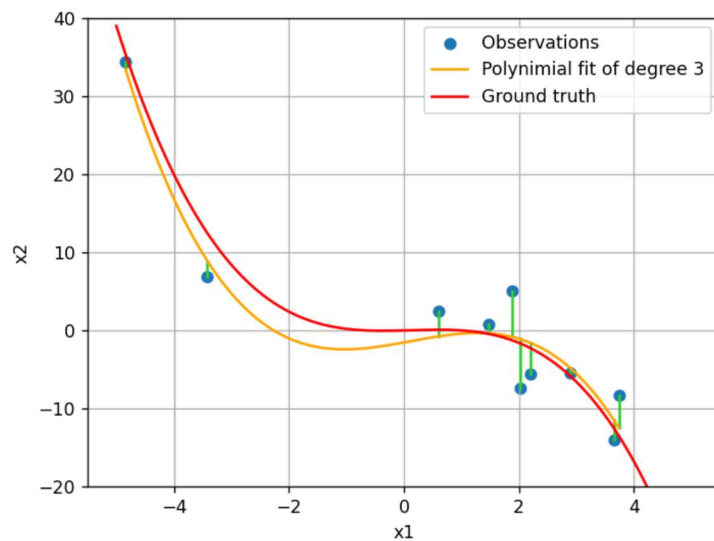
MSE=34.222

With a degree of 1, the polynomial tells us the general direction of the data (AKA the linear regression)



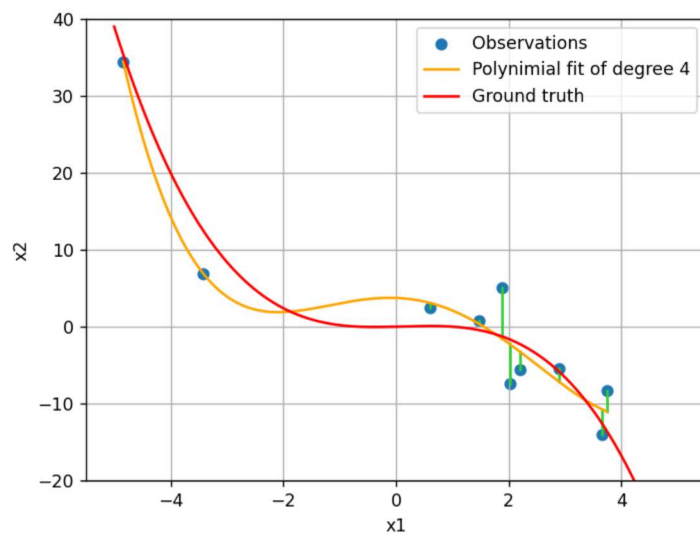
MSE=32.958

With a degree of 2, the polynomial is still underfitted



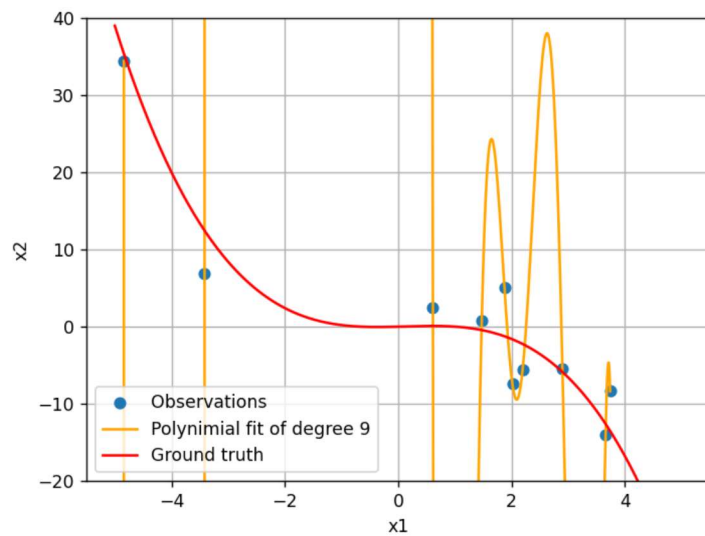
MSE=12.963

This is the polynomial closest to the ground truth, and is neither overfitted nor underfitted



MSE=9.645

At 4 degrees, the polynomial fits rather well, and still yields an idea of what the data could tell us

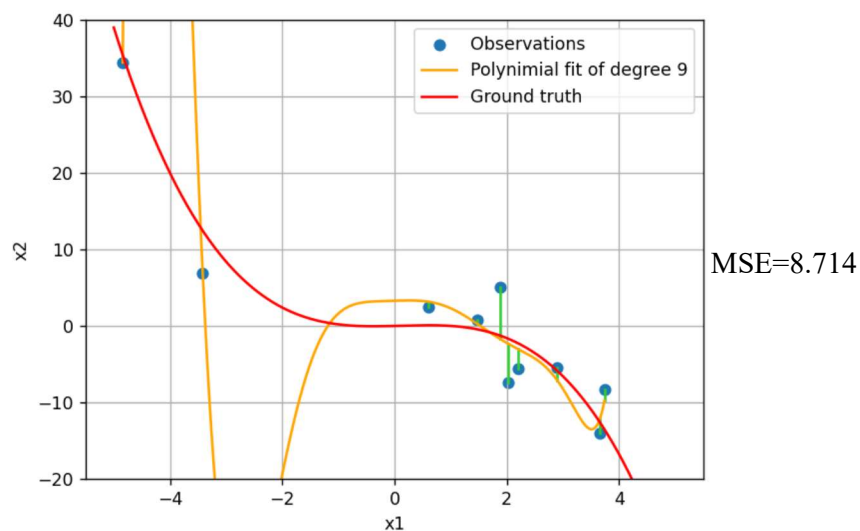


MSE=0

As can be seen, even though the MSE is at 0 now, the polynomial is overfitted and therefore cannot tell us anything about actual data.

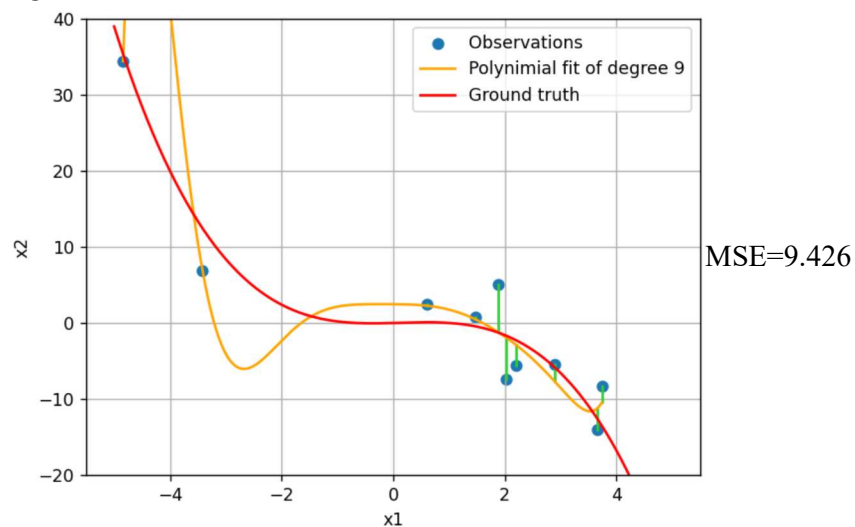
Exercise 2

l_2 penalization with $\text{reg}=1.01$:



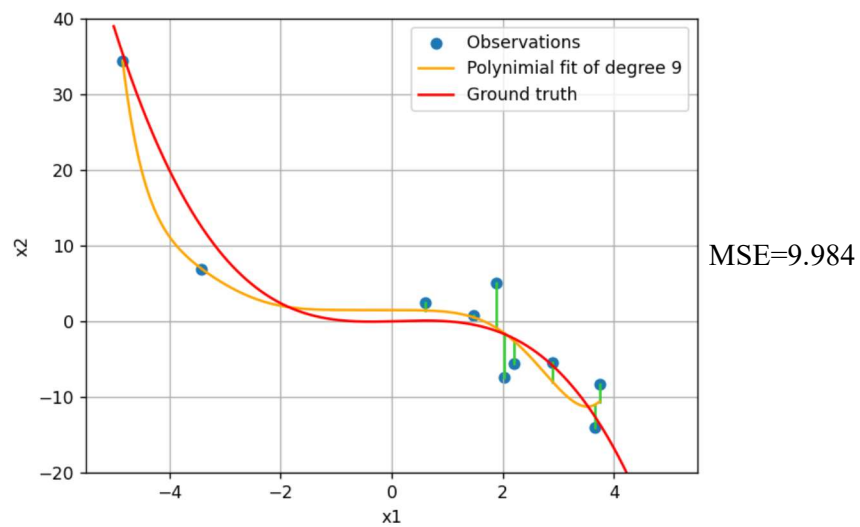
With the added penalization, the polynomial is closer to the actual data , but still far from the ground truth.

reg=10.001:



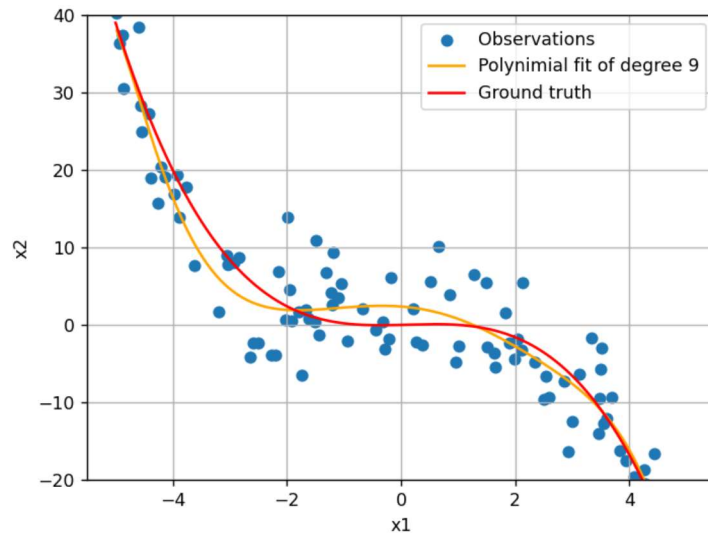
As the penalty has increased 10-fold the the polynomial begins to align better with the ground truth

reg=100.0001



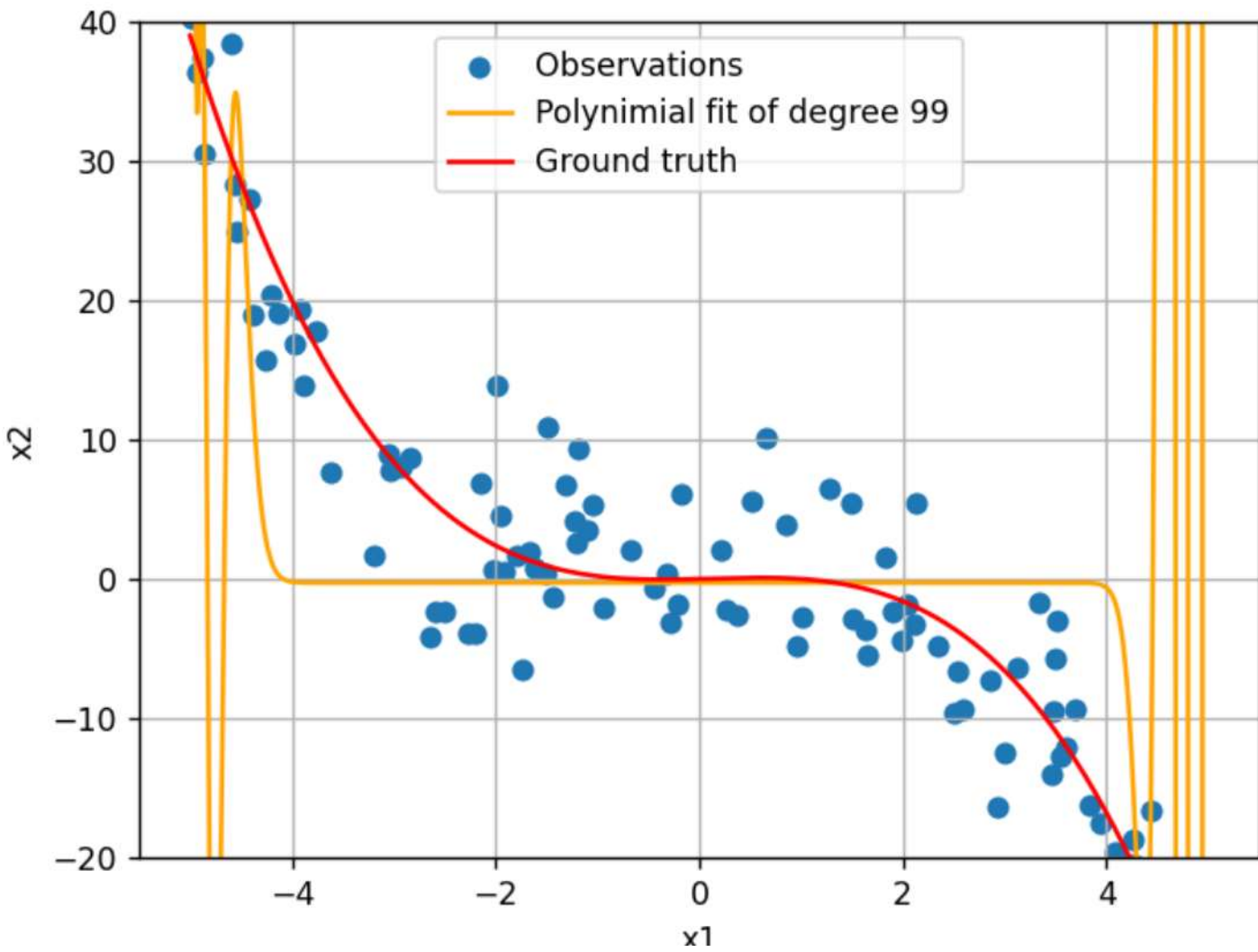
And with another 10-fold penalty, the ground truth is rather close, but still affected by the small sample size.

Exercise 3



MSE=18.91

As can be seen from the plot, the added number of samples has in this specific instance not made much of a difference. However, for this, the regularization is completely neglected, but still yields results similar to the regularizer set at 100. This shows that an added amount of samples will definitely have an effect on the polynomial fit.



For fun, if the polynomial is increased to 99, the polynomial is both underfitted (for most of the data) and overfitted (for some specific values). Which just shows that there are not set polynomial or sample number, and it all depends on the data at hand.